

06.05 Factors and Rational Roots

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
Polynomial Structure And Equations	Look first for structure: degree, factor pattern, roots, or graph behavior.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

A. Practice: Factors and Rational Roots

1.

$$x^2 + 7x + 12$$

2.

$$2x^2 - 5x - 3 = 0$$

3.

$$f(2) = 2^3 - 2(2) + 5$$

4.

$$x^2 + 7x + 12$$

5.

$$2x^2 - 5x - 3 = 0$$

6.

$$f(2) = 2^3 - 2(2) + 5$$

7.

$$x^2 + 7x + 12$$

8.

$$2x^2 - 5x - 3 = 0$$

9.

$$f(2) = 2^3 - 2(2) + 5$$

10.

$$x^2 + 7x + 12$$

11.

$$2x^2 - 5x - 3 = 0$$

12.

$$f(2) = 2^3 - 2(2) + 5$$

13.

$$x^2 + 7x + 12$$

14.

$$2x^2 - 5x - 3 = 0$$

15.

$$f(2) = 2^3 - 2(2) + 5$$

16.

$$x^2 + 7x + 12$$

17.

$$2x^2 - 5x - 3 = 0$$

18.

$$f(2) = 2^3 - 2(2) + 5$$

19.

$$x^2 + 7x + 12$$

20.

$$2x^2 - 5x - 3 = 0$$

B. Apply the key formulas

1.

$$a^2 - b^2 = (a - b)(a + b)$$

2.

$$b = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

3.

$f(a) = 0 \Rightarrow (x - a)$ is a factor

4.

$$a^2 - b^2 = (a - b)(a + b)$$

5.

$$e = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

6.

$f(a) = 0 \Rightarrow (x - a)$ is a factor

7.

$$a^2 - b^2 = (a - b)(a + b)$$

8.

$$c = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

9.

$f(a) = 0 \Rightarrow (x - a)$ is a factor

10.

$$a^2 - b^2 = (a - b)(a + b)$$

11.

$$a = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

12.

$f(a) = 0 \Rightarrow (x - a)$ is a factor

13.

$$a^2 - b^2 = (a - b)(a + b)$$

14.

$$d = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

15.

$f(a) = 0 \Rightarrow (x - a)$ is a factor

16.

$$a^2 - b^2 = (a - b)(a + b)$$

17.

$$b = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

18.

$f(a) = 0 \Rightarrow (x - a)$ is a factor

19.

$$a^2 - b^2 = (a - b)(a + b)$$

20.

$$e = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$